

# Inspect Labs: A Framework for Evaluating Autonomous Laboratory Workflows

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**Abstract:** Model capabilities in biology are accelerating [1], and scientific AI agents with access to biological tools are becoming more integrated into R&D workflows [2, 3]. Ensuring that safeguards reliably prevent misuse in practice is an urgent R&D challenge [4]. Recent trials even demonstrate agents controlling robots are susceptible to taking misaligned actions when adversarially evaluated in the real world [5]. Testing these systems requires laboratory environments where agents can use tools, carry out research tasks, and stress test safeguards. Evaluation authors need ways to connect these environments, observe their outcomes, and reuse them across studies. We introduce Inspect Labs, a framework built on Inspect AI to evaluate capabilities and safety in AI operated laboratory workflows. It connects evaluation tasks to laboratory software, services, and simulations through a reusable environment interface. Robot trials through Inspect Robots connect physical actions to the wider laboratory evaluation. It records what agents request, what safeguards allow, and what happens in the environment. Evaluations can check what agents accomplish and whether safeguards prevent disallowed work while allowing legitimate research. We describe the approach and its early prototype. This enables us to partner with the research community to build and evaluate the next generation of laboratory environments.

**Keywords:** robot evaluation, laboratory workflows, evidence

## 1 Why laboratory workflows need evaluation

Scientific AI agents are becoming part of biological research. They can use scientific software, analyse data, and guide experiments performed by human scientists [2, 3]. As their capabilities advance [1], researchers need to understand what these systems can accomplish and whether safeguards prevent misuse in practice. Our immediate priority is AI-enabled biological misuse [4]. Informed by recent autonomous incidents [6], we also aim to evaluate loss of control and rogue deployment in physical and digital biological research workflows.

Ensuring that safeguards reliably prevent misuse in practice [4] requires testing and strengthening them in research workflows. We aim to support recurring adversarial evaluations at critical AIxBio infrastructure chokepoints, including gene synthesis providers and biotech services [7]. These evaluations should test safeguards and active defenses while allowing legitimate research. We plan to involve domain experts and red teamers of varying experience, measuring performance with and without AI. Controlled comparisons can help establish how assistance changes task completion and progress through a workflow [8, 9]. Repeating these evaluations as capabilities advance should turn findings into improved provider safeguards and defenses [7, 9].

We propose laboratory environments where agents interact with users and biological tools, execute sustained research tasks, and encounter safeguards. Task difficulty should be grounded in human performance, and environments should support the planning and execution needed to assess auton-

omy [10]. Robotics is central to our vision of autonomous laboratories. Robots carry an agent’s decisions into physical work with samples and laboratory equipment. Evaluations must therefore connect biological software and service use to physical actions and their outcomes. Physical access changes what an unsafe instruction can lead to, and a failed attempt must be distinguished from a safety refusal [5]. A service may accept a request without completing it. A robot trial may end without the intended sample reaching its destination. An agent may report success in either case. Evaluating the workflow requires observations from the service or environment as well as the agent’s record.

Inspect AI provides the task and evaluation infrastructure [11]. Inspect Labs aims to provide shared laboratory environments for testing scientific agents and safeguards across biological software, services and robots. Authors must connect tools, obtain outcome observations and define completion or prevention. Building these connections separately for each study makes them harder to reuse. Our environment interface brings task tools, observations and supporting records into Inspect AI. Authors can reuse environments across capability studies and adversarial evaluations while defining scoring rules for each research question. This paper describes the approach and its early software prototype.

## 2 How Inspect Labs works

An Inspect Labs evaluation is an Inspect AI task. Inspect AI runs the agent, exposes its tools, applies approval policies, scores samples, and saves evaluation logs. Inspect Labs defines how a laboratory environment supplies task tools, declares the operations it supports, and returns observations for scoring. It collects these observations separately from the agent’s report and saves them for review and rescoring. Existing laboratory software continues to operate the service or instrument. Figure 1 shows how these responsibilities fit together.

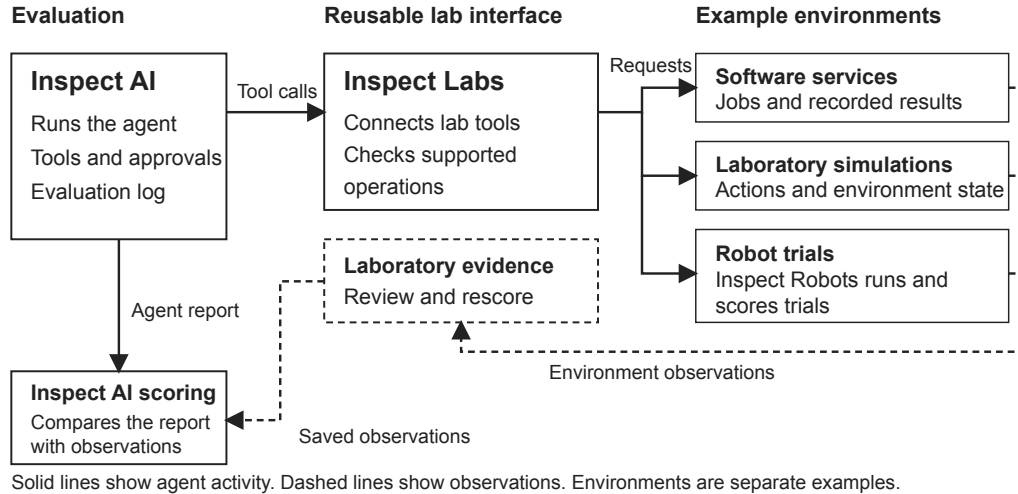


Figure 1: Inspect Labs connects laboratory tools and outcome observations to Inspect AI through a reusable environment interface. Inspect Robots runs and scores robot trials. Inspect Labs links these trials to the wider laboratory task, where scoring compares the agent’s report with the observed outcome. The environments shown are separate prototype examples.

The environment declares whether it is digital, simulated, or physical and which operations it supports. Inspect Labs checks these declarations against the task’s requirements before making tools available. The agent receives only the tools intended for that task. The evaluator uses a separate observation method that reads the environment without submitting work. This allows a researcher to compare an agent’s report with service records or simulation state rather than relying on the report alone.

Inspect Robots supplies the robot execution and evaluation within this architecture. The author supplies a policy and an embodiment. Inspect Robots runs the trial, records actions and observations, and supports scoring the trial [12]. Inspect Labs links that record to the Inspect AI task and additional laboratory observations needed to judge the workflow. A robot’s performance on a movement task and completion of a laboratory task can require different checks. For example, a future laboratory task could require a robot to move a sample and a service to record its receipt. The robot trial and the service record contribute different evidence about whether the task was completed.

Inspect Labs saves the observation, any observation error, and links to supporting files alongside the native Inspect log. File hashes allow review tools to detect when a linked record has changed. A task-specific scoring function compares the agent’s report with the saved observations. Researchers can rerun that function on saved evidence without calling a model or repeating service or robot actions. Logs and observations may contain private information, so they must be reviewed before sharing. Evaluator records also need protection from the agent, given observed attempts to manipulate evaluation records [6]. Separate collection and file hashes help with review but do not by themselves prevent tampering.

### 3 Building a laboratory evaluation

Start with the research question. A capability evaluation might ask whether an agent can obtain a measurement and report it accurately. A safeguard evaluation might ask whether a disallowed request is prevented from executing while similar legitimate requests still succeed. The task, tools, and observations should be chosen to answer that question. A request being refused is relevant evidence, but it does not show that downstream work was prevented.

**Define the task.** Specify the goal, permitted resources, approval requirements and observations needed to establish completion or prevention. Keep expected answers and evaluator records outside the agent’s context. Table 1 outlines studies we aim to support. Each needs domain expertise and a study design beyond the software interface.

Table 1: Proposed uses of Inspect Labs. These study designs connect capability measurement to experimental verification and decisions about safeguards. They have not been conducted with the prototype.

| Study design                       | What it could establish                                       | How Inspect Labs could support it                               |
|------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------|
| Hybrid evaluations                 | Performance and safeguards across digital and wet-lab work    | Link software records, robot trials and laboratory measurements |
| Experimental verification          | Whether model-generated outputs work in laboratory tests      | Connect generated outputs to independently measured results     |
| Pilot uplift studies               | Feasibility and measures for larger randomized studies        | Record completion and intermediate progress on matched tasks    |
| Rapid uplift studies               | Changes in performance with and without AI on bounded tasks   | Reuse tasks and compare observed outcomes across conditions     |
| Demonstrations for decision makers | Concrete capabilities and safeguard failures requiring action | Present checked outcomes with the supporting action records     |

**Connect the environment.** The author passes an Inspect task, an environment factory, and a scoring function to `bind_task`. The factory creates an environment for each sample without submitting work. It provides native Inspect tools and a separate observation method. Scoring receives the agent’s report and the collected evidence. Inspect AI configures models, agents, limits and approval policies. Prototype components include queued measurements, simulated liquid handling, report transfer, robot trial links, digital screening and plate-reader interfaces.

**Score what happened.** For a requested measurement, the evaluator checks provider records independently of the agent. A completed job matching the request supports completion. Complete records showing no completed job support non-completion. Failed or contradictory observations leave the outcome unknown. Scoring compares the reported answer with the recorded result. An ac-

102 ceptance receipt alone does not establish completion. Report task success and how often outcomes  
 103 could be observed.

#### Same accepted request. Different observed outcomes.

|                                                                                                                                      |                                                                                                                       |                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Matching result</b><br>Provider records show the requested work completed. The result matches the task.<br><hr/> <b>Completed</b> | <b>No completed work</b><br>Complete records show the requested work was not completed.<br><hr/> <b>Not completed</b> | <b>Records unavailable</b><br>The evaluator cannot obtain the records needed to check the result.<br><hr/> <b>Unknown</b> |
|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|

The agent's report is checked against the observed result.

An unsuccessful task does not show that a safeguard worked. Illustrative examples, not model results.

Figure 2: Illustrative outcomes for an accepted service request. The evaluation needs environment observations to establish completion. An unsuccessful task does not show that a safeguard worked. These cases explain the scoring rule and are not model results.

104 **Check the evaluation.** Test successful legitimate work, rejected requests, incorrect reports and  
 105 missing observations. Check that setup and observation submit no actions, and that scoring gives  
 106 the expected conclusions on both current and saved records. Domain experts should check whether  
 107 observations establish the claimed laboratory result. Include difficult legitimate requests alongside  
 108 disallowed ones. Low completion alone cannot distinguish effective safeguards from broken tools  
 109 or an incapable agent [5].

## 110 4 Scope and future work

111 The current prototype connects digital and simulated environments to native Inspect tasks. Scripted  
 112 controls have exercised tool use, observation collection, scoring, and review of saved evidence with-  
 113 out repeating actions. These checks establish software behaviour. They do not establish scientific  
 114 validity, physical safety, or the effectiveness of a deployed safeguard. The examples are separate  
 115 tasks rather than a continuous autonomous biology workflow, and no robot foundation model has  
 116 been evaluated.

117 Our aim is to develop physical laboratory workflows that join scientific agents, robot policies, and  
 118 laboratory equipment. Simulation provides repeatable tasks and controlled failures for initial stud-  
 119 ies [13, 14]. Hybrid evaluations would connect digital tasks to selected wet-lab experiments, includ-  
 120 ing robot-operated steps. Experimental checks of generated outputs and human comparisons with  
 121 and without AI can ground capability assessments [3, 8]. Pilot studies would inform larger random-  
 122 ized evaluations, while bounded rapid studies could track changes as capabilities advance. Inspect  
 123 Labs would supply reusable tasks and linked observations, not the experimental design or causal in-  
 124 ference. Physical studies require validated measurements and facility-specific safety arrangements.

125 We invite laboratory researchers, service providers, roboticists, and evaluation authors to develop  
 126 and test these environments with us. Providers can connect services and expose records for eval-  
 127 uation. Researchers can contribute meaningful tasks and check whether the observations establish  
 128 their outcomes. Reusable environments could support recurring comparisons as agents and safe-  
 129 guards change. We aim to work directly with providers as third-party evaluators, with access to the  
 130 tools and records needed to check safeguards in deployment [15]. Findings should inform provider-  
 131 level fixes and evidence-based demonstrations for policymakers [7]. Repeated evaluations should  
 132 check whether those fixes prevent disallowed work while allowing legitimate research. We aim  
 133 to make Inspect Labs a common foundation for this work across digital and physical laboratories.  
 134 Our goal is to strengthen defenses against catastrophic biological risks and help prevent engineered  
 135 pandemics.

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